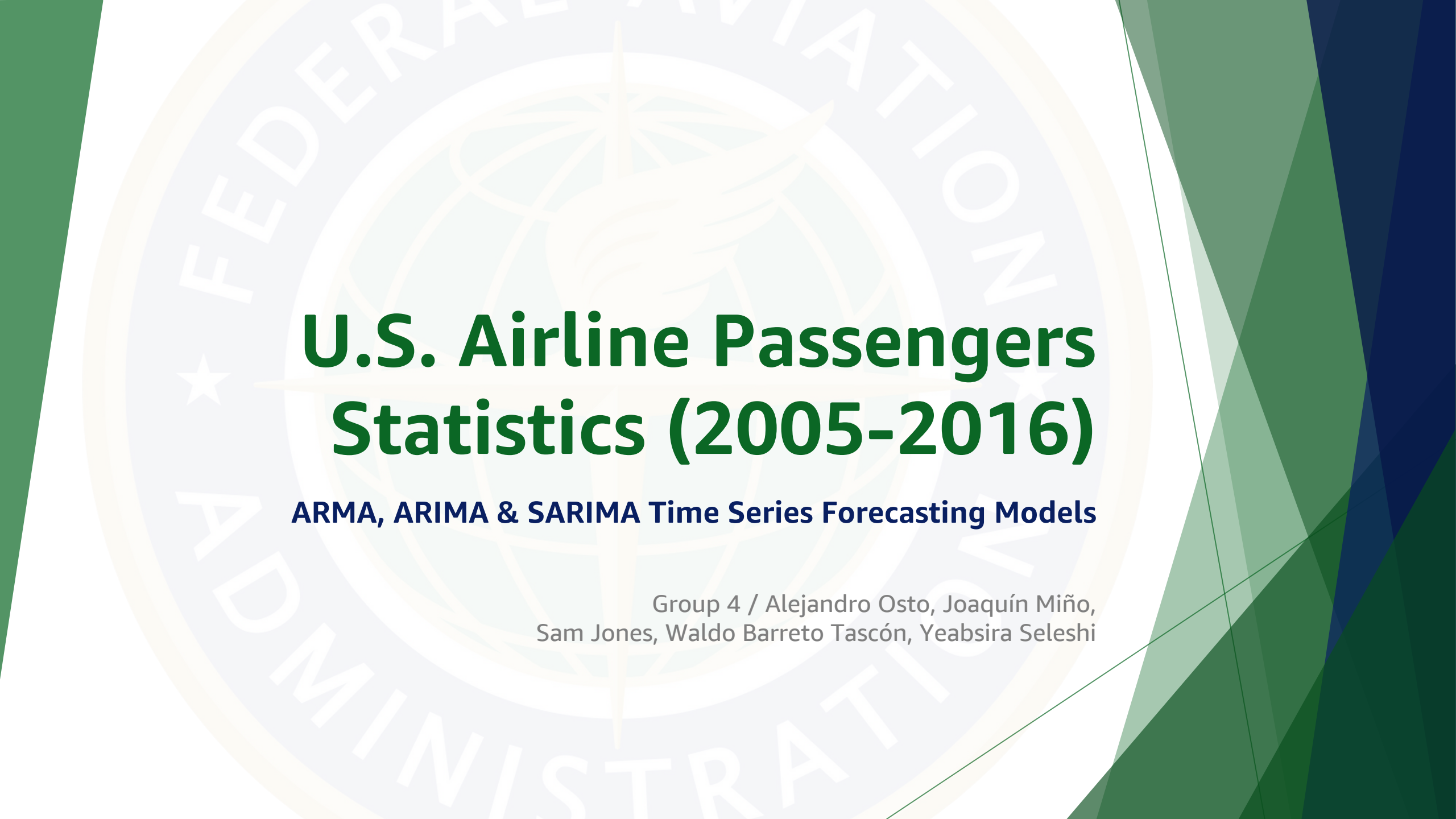




U.S. Airline Passengers Statistics (2005-2016)

ARMA, ARIMA & SARIMA Time Series Forecasting Models

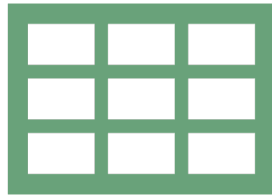
Group 4 / Alejandro Osto, Joaquín Miño,
Sam Jones, Waldo Barreto Tascón, Yeabsira Seleshi

Dataset Overview

U.S. Airlines Traffic Passenger Statistics



Time Range
2005-2016



15007 Rows
17 Columns



International **61%**
Domestic **39%**



USA **39%**
Asia **22%**
Other **39%**



UNITED

United Airlines
20%



Dataset Transformation

Raw

1

	index	Activity Period	Operating Airline	Operating Airline IATA Code	Published Airline	Published Airline IATA Code	GEO Summary	GEO Region	Activity Type Code	Price Category Code	Terminal	Boarding Area	Passenger Count	Adjusted Activity Type Code
0	0	200507	ATA Airlines	TZ	ATA Airlines	TZ	Domestic	US	Deplaned	Low Fare	Terminal 1	B	27271	Deplaned
1	1	200507	ATA Airlines	TZ	ATA Airlines	TZ	Domestic	US	Enplaned	Low Fare	Terminal 1	B	29131	Enplaned
2	2	200507	ATA Airlines	TZ	ATA Airlines	TZ	Domestic	US	Thru / Transit	Low Fare	Terminal 1	B	5415	Thru / Transit * 2
3	3	200507	Air Canada	AC	Air Canada	AC	International	Canada	Deplaned	Other	Terminal 1	B	35156	Deplaned
4	4	200507	Air Canada	AC	Air Canada	AC	International	Canada	Enplaned	Other	Terminal 1	B	34090	Enplaned

2

```
df['Year'] = df['Activity Period'].astype(str).str[0:4]
df['Month'] = df['Activity Period'].astype(str).str[4:]

df['Period'] = df['Month'] + '-01' + '-' + df['Year']
df['Period'] = pd.to_datetime(df['Period'])
df['YearDay'] = df['Period'].dt.dayofyear

flights = pd.DataFrame(df.groupby('Period')['Passenger Count'].sum())
```

Transformations & Group By

3

Passenger Count	
2005-07-01	3225769
2005-08-01	3195866
2005-09-01	2740553
2005-10-01	2770715
2005-11-01	2617333

Time Series



Time Series Overview

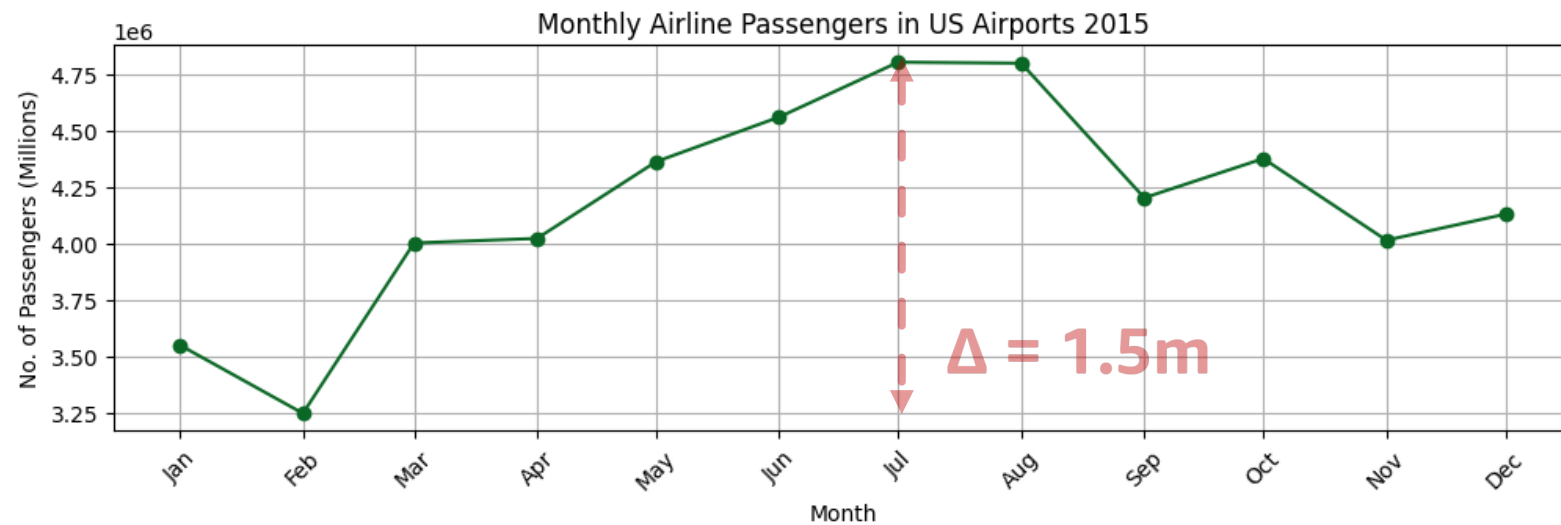
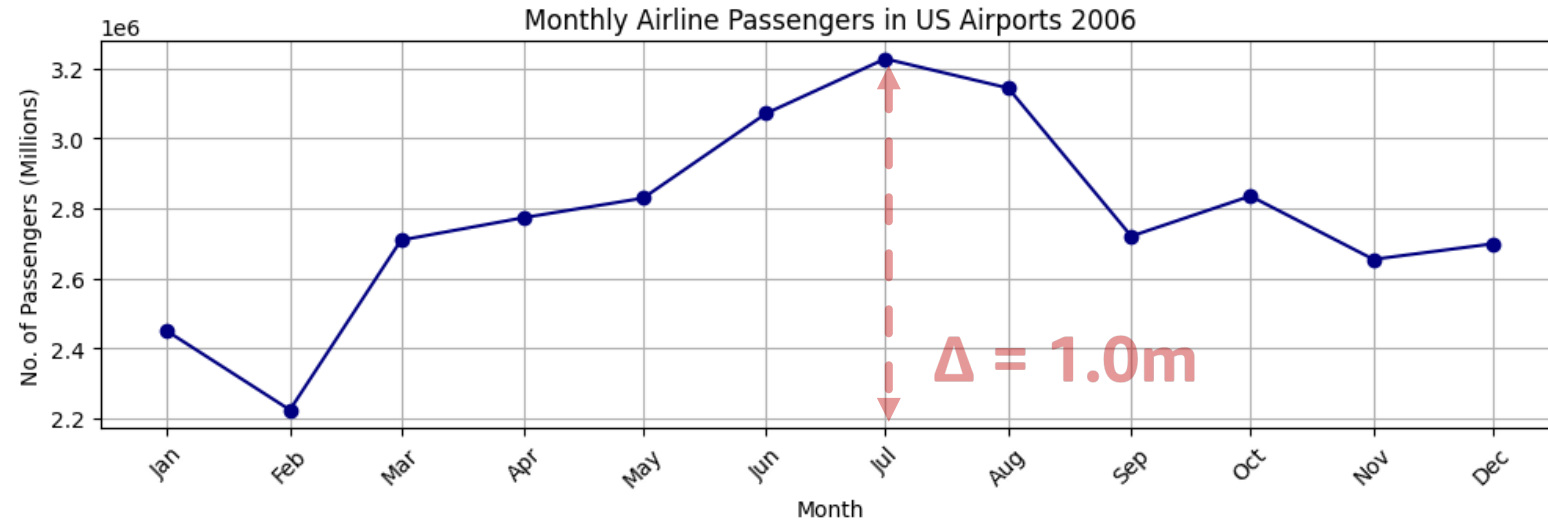


Seasonality

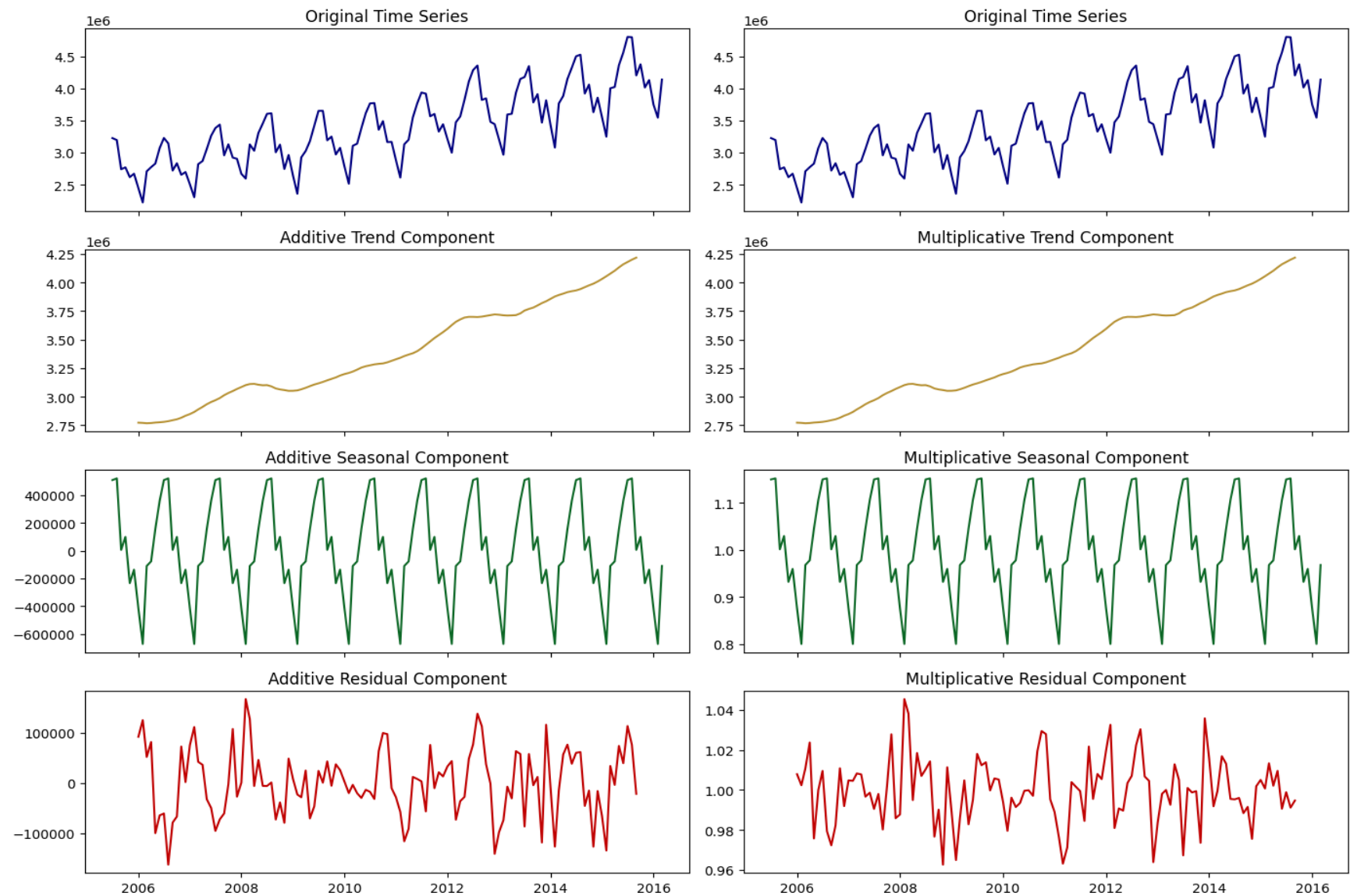
2006

VS.

2015

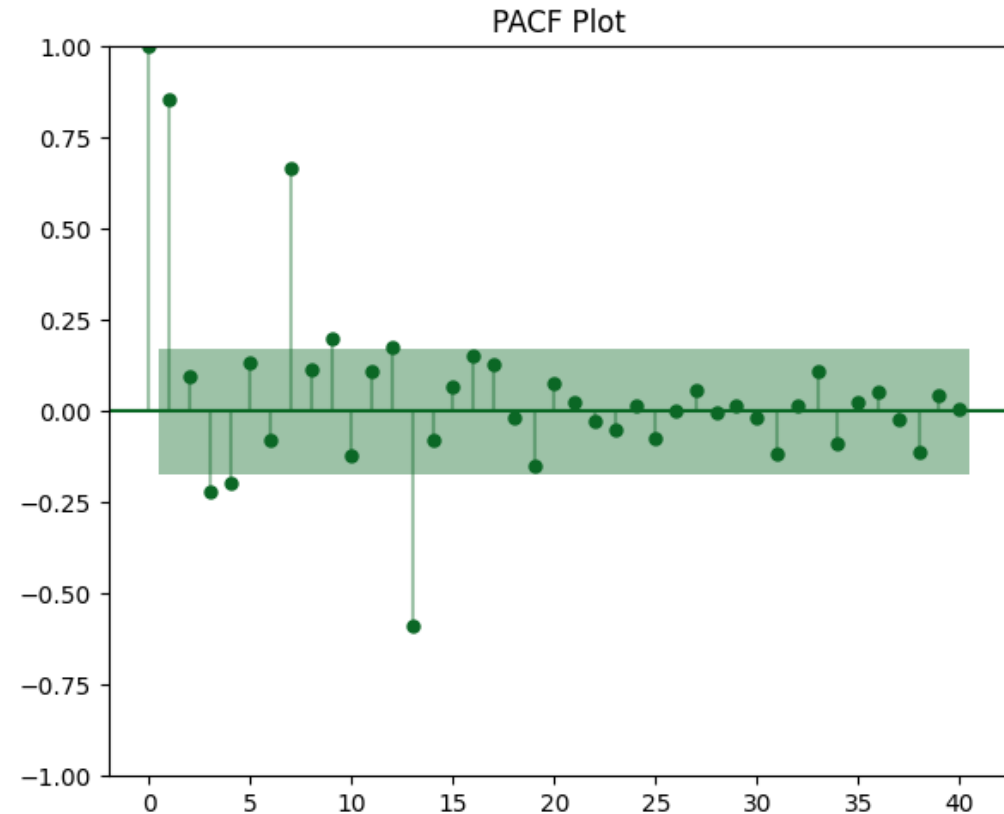
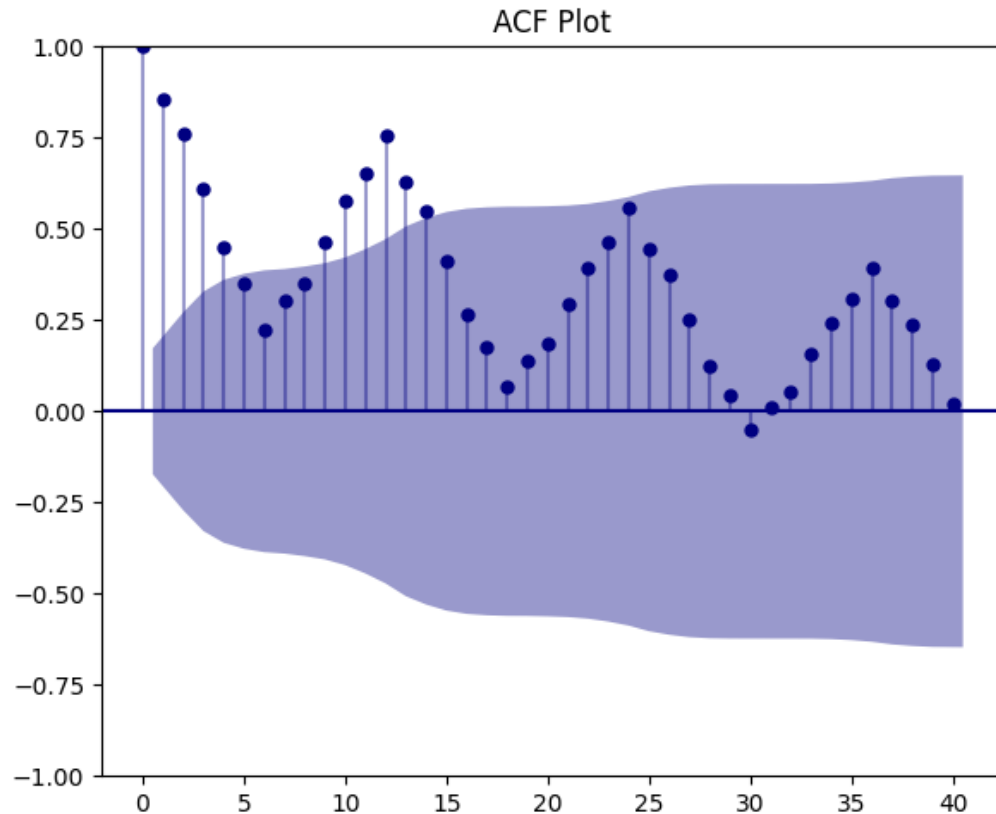


Decompositions



ACF/PACF

Before Differencing

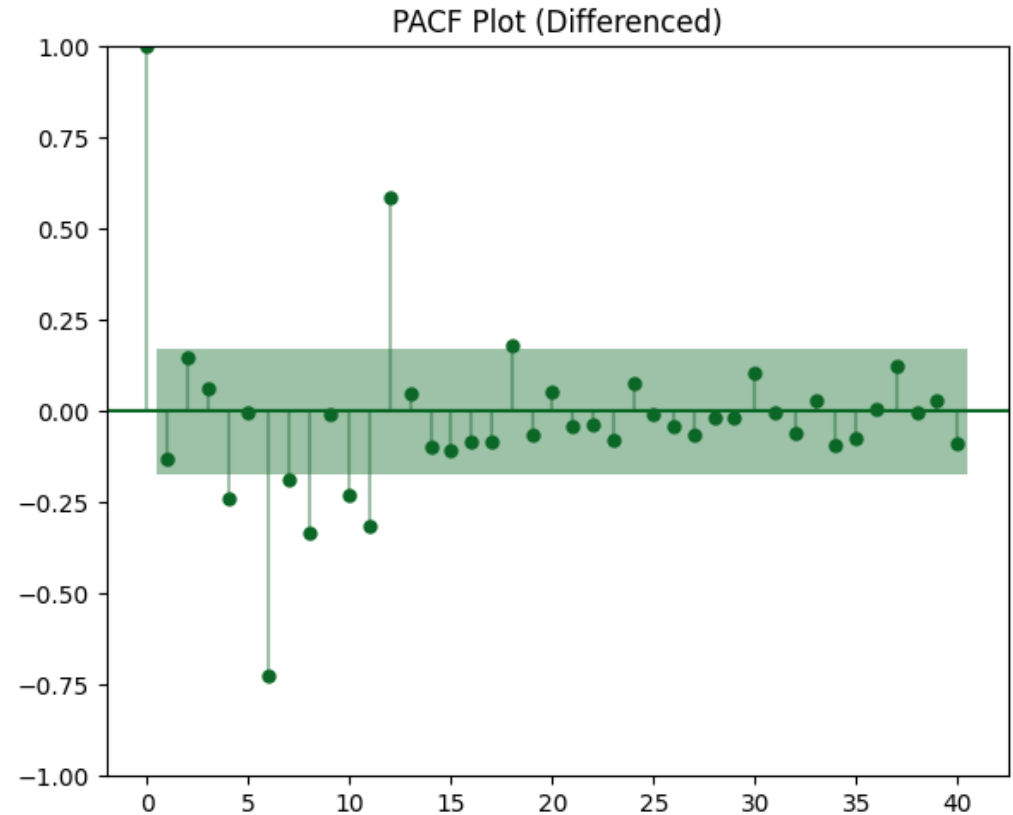
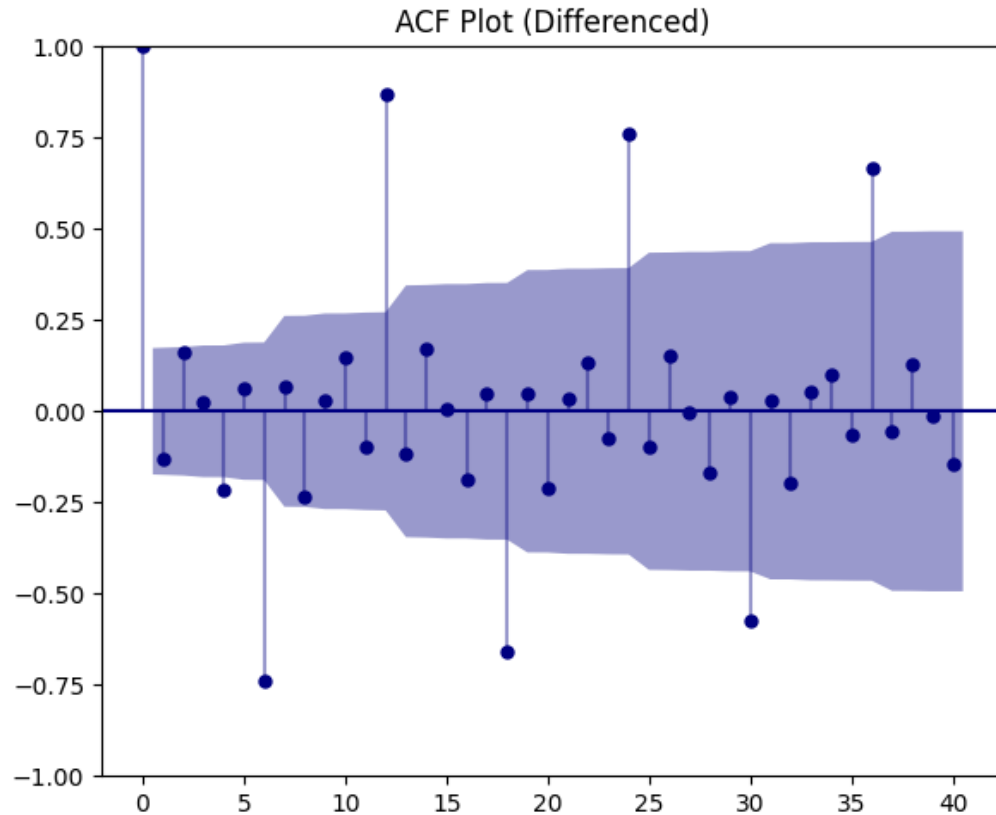


Evidence of seasonality in the ACF and PACF with the Lag12 values



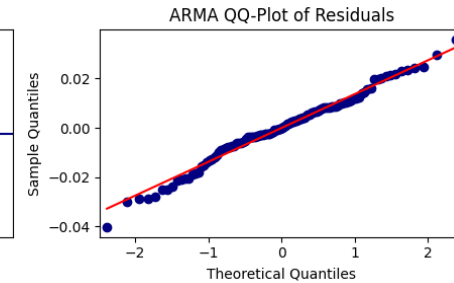
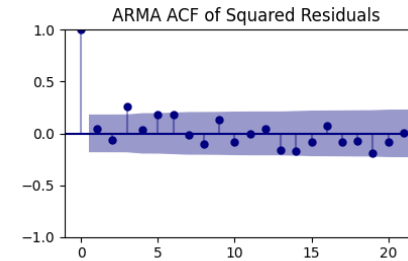
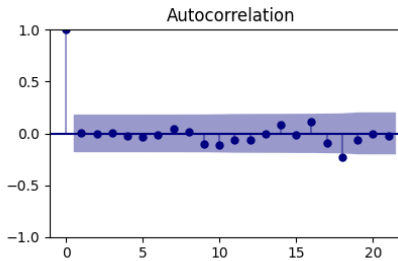
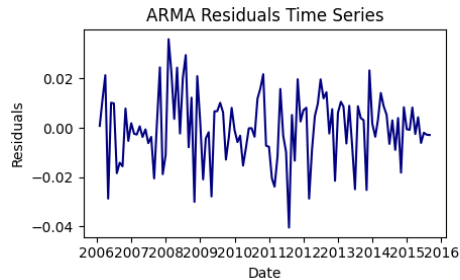
ACF/PACF

After Differencing (d=1)

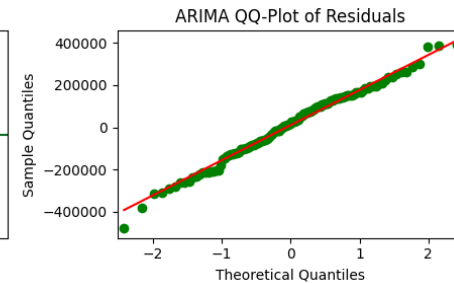
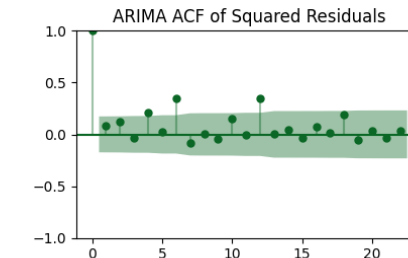
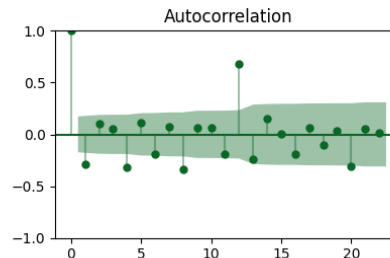
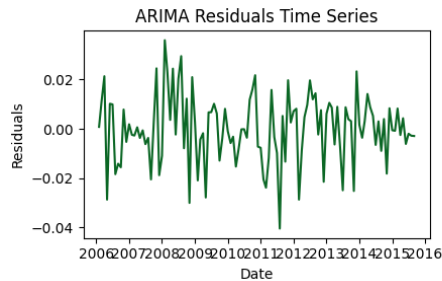


Residuals per Model

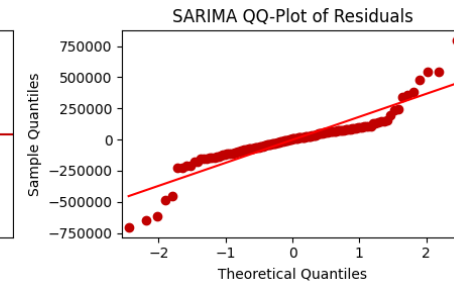
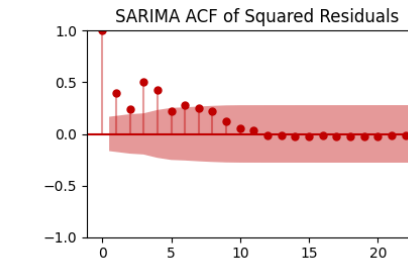
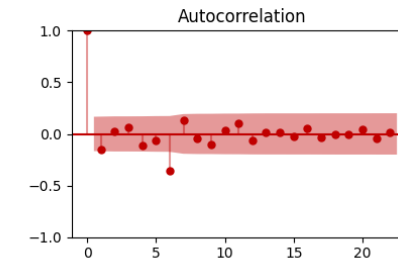
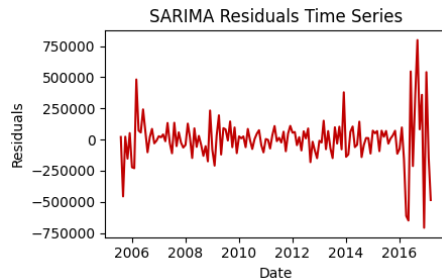
ARMA



ARIMA

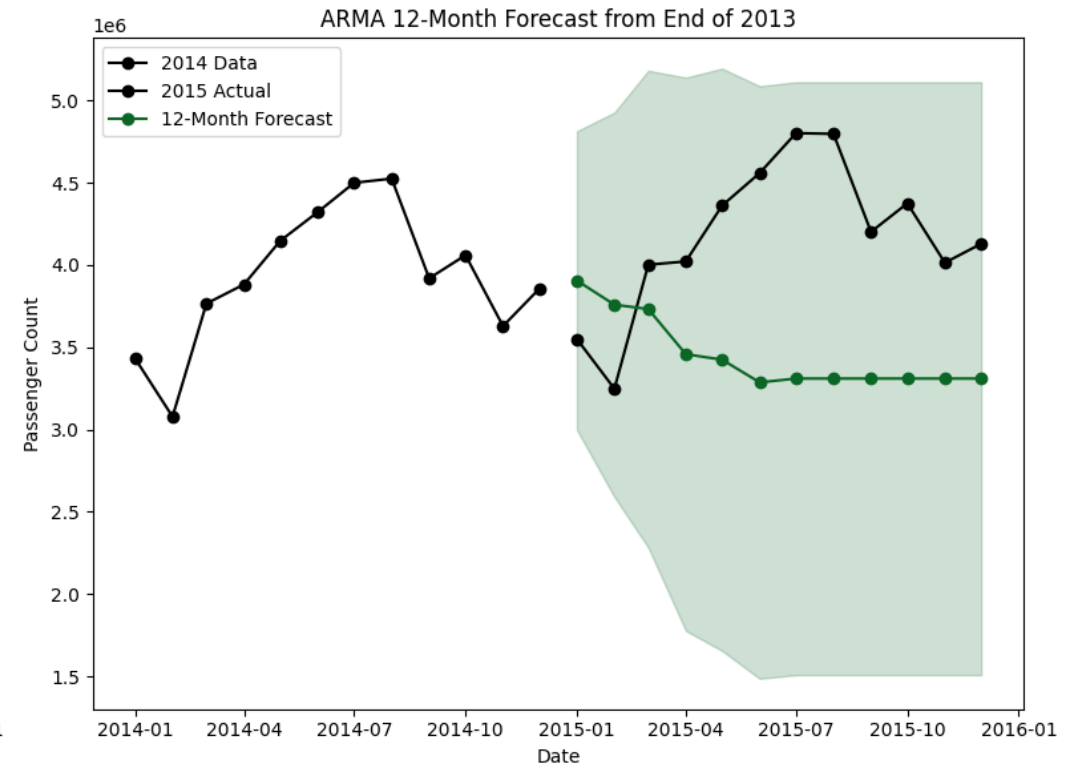
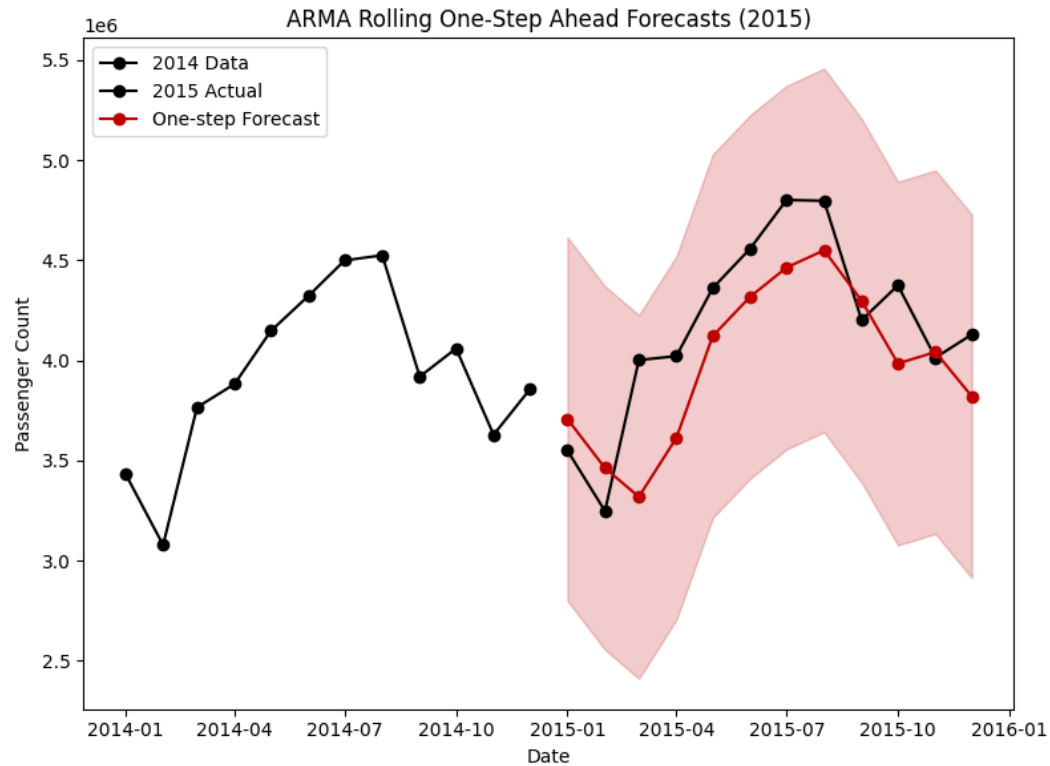


SARIMA



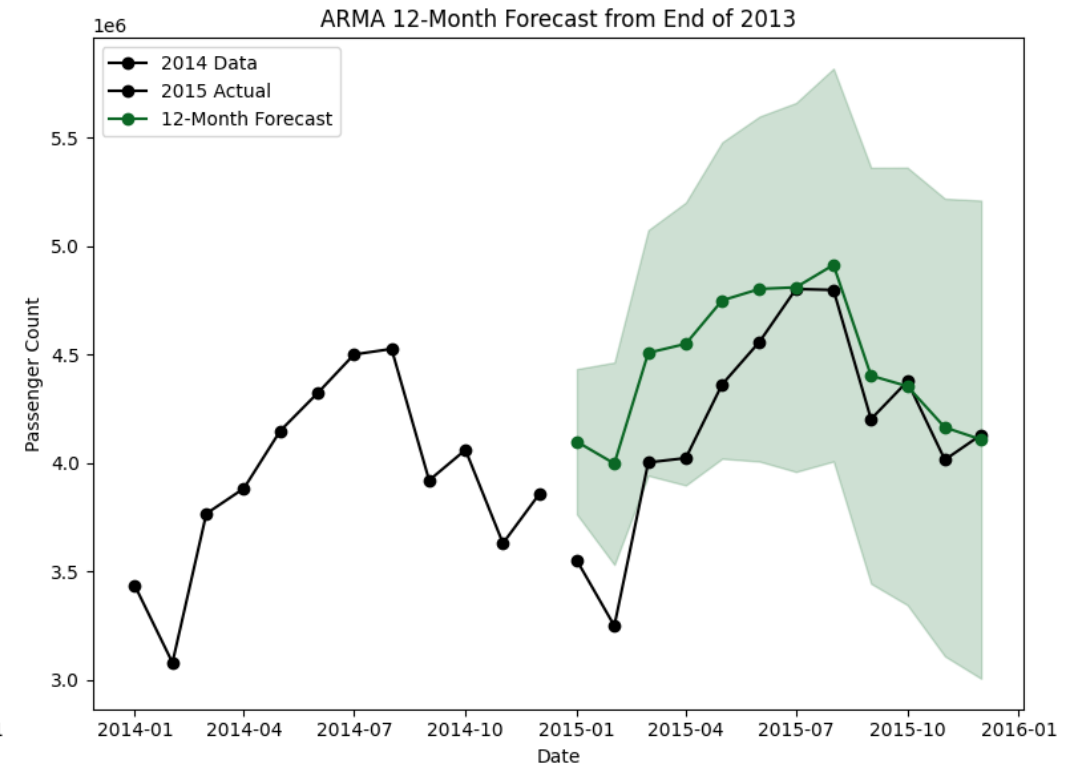
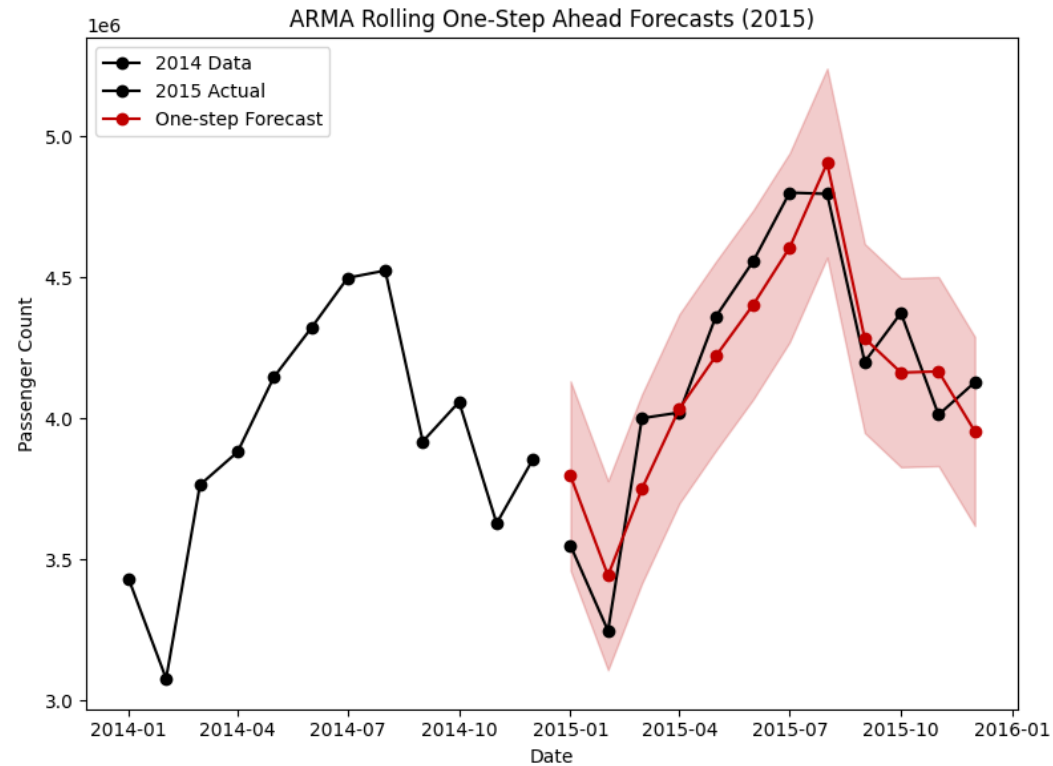
Forecasting Performance

ARMA (with decomposition residuals)



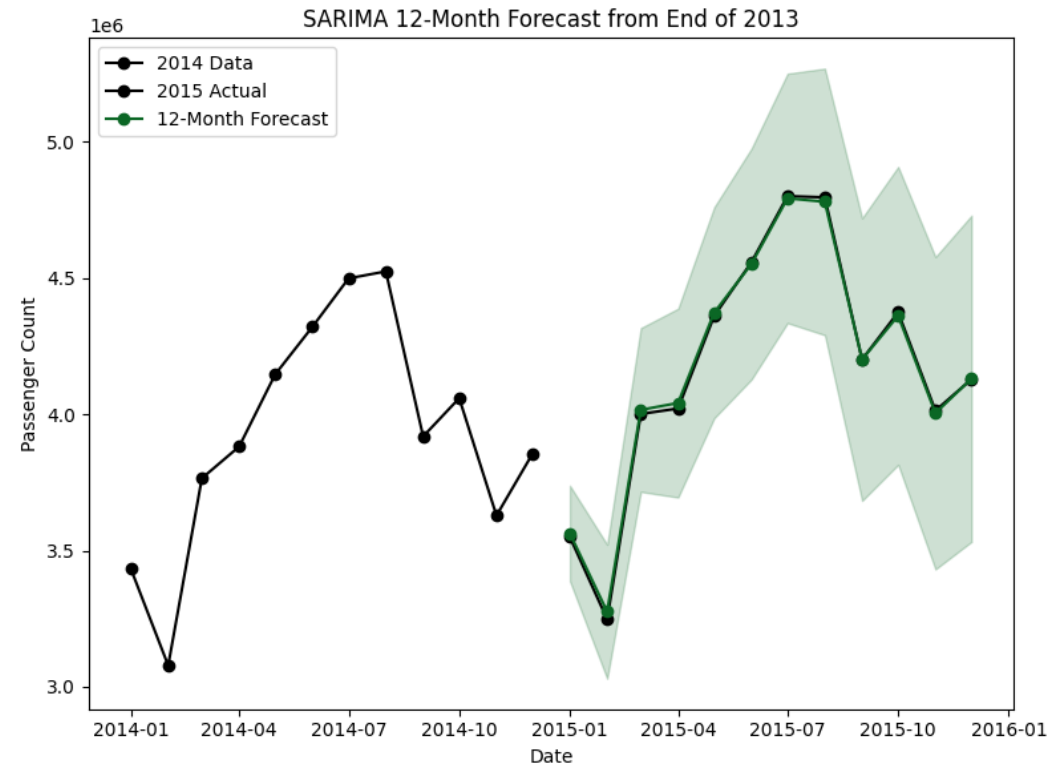
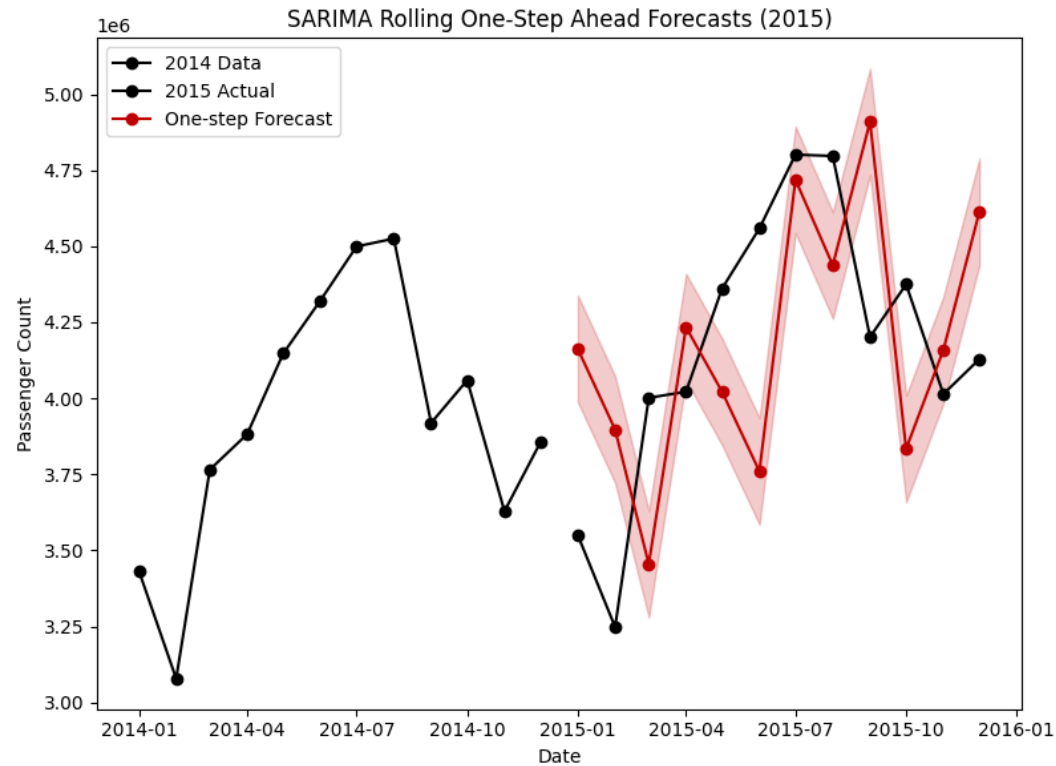
Forecasting Performance

ARIMA



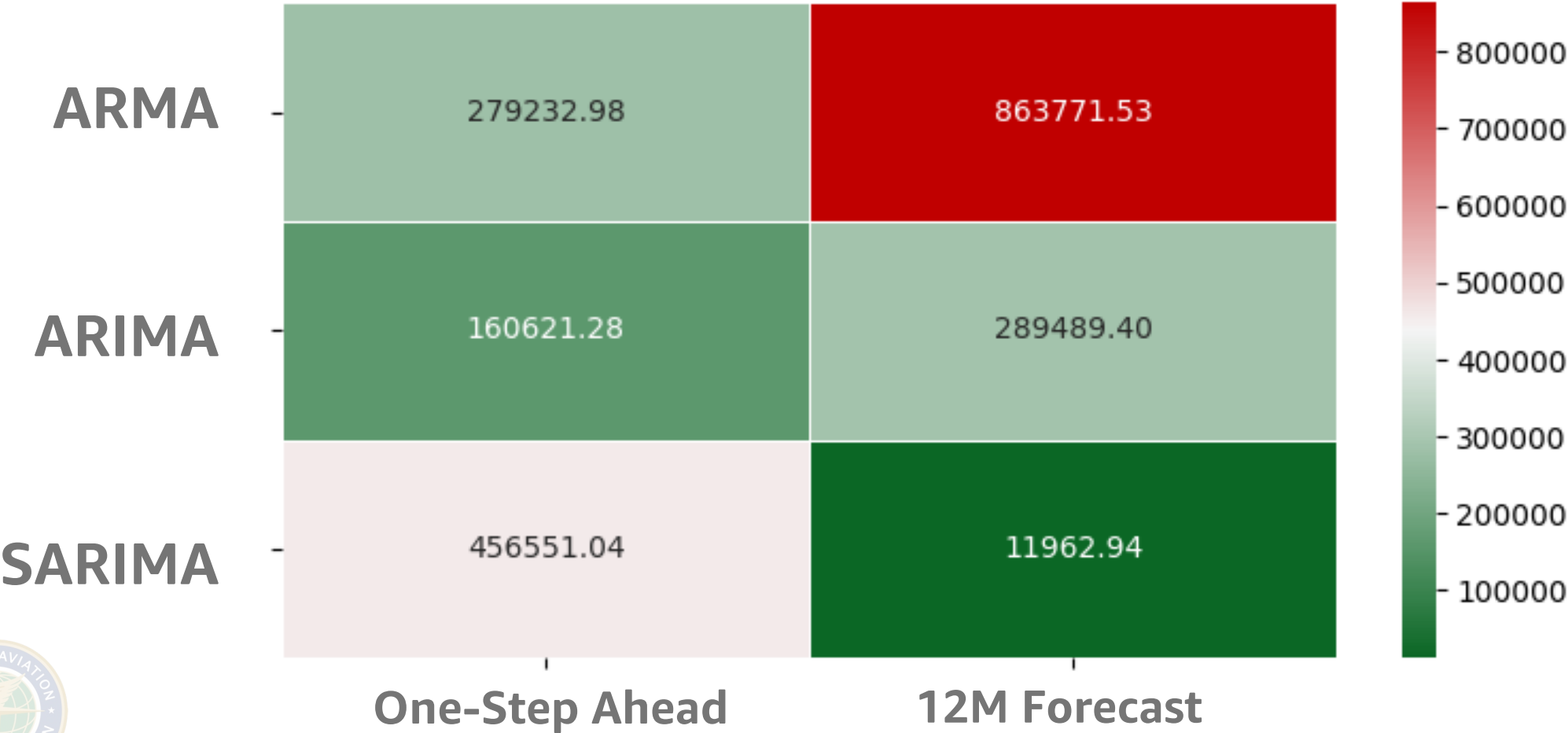
Forecasting Performance

SARIMA



Model Comparison

Mean Absolute Error



Key Insights & Implications

- Seasonality plays a major role in fluctuations, indicating higher demand in certain months.
- **Multiplicative Decomposition** was required due to the variability within the seasonal component over time (non-constant seasonality) as the aviation industry grows. This is generally more accurate for economic and transport data.
- **In Augmented Dickey-Fuller (ADF) test:**
 - Pre-differencing: ADF Statistic = 0.2342, P-value = 0.9741 → Non-stationary
 - Post-differencing: ADF Statistic = -3.2112, P-value = 0.0193 → Stationary
- **ARIMA** was our model of choice due to lowest MAE and best fit to line in residual graphs
- **Real-world applications of the model include:**
 - Adjusting aircraft allocations based on predicted demand.
 - Expanding facilities during high-traffic seasons.
 - Optimizing ticket pricing strategies.
 - **Challenges & limitations:**
 - External factors like fuel prices, pandemics, and economic downturns can affect accuracy.
 - Assumption of past trends continuing in the future may not always hold.

Overall, future forecasting models can help airlines plan better in terms of capacity, pricing, and operational efficiency.

